

Arun Prashath

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SKILLS

Languages: Python, C++, C, SQL, Java
Tools/Platforms: Pandas, NumPy, Scikit-learn, Prophet, Streamlit, SQLite, Git, GitHub
Soft Skills: Team Player, Time Management, Attention to Detail

PROJECTS

Student Performance Prediction Using Machine Learning | Python, Pandas, NumPy, Scikit-learn May' 26

- Trained and benchmarked a **supervised regression** pipeline predicting students' final grade from **649 records across 30+ attributes**, supporting early identification of students needing intervention.
- Cleaned and processed data through **IQR outlier treatment**, categorical **encoding** and leakage-free **scaling**; evaluated **Linear Regression**, **Random Forest** and **PCA + Linear Regression** on an 80:20 split with **5-fold cross-validation**.
- Achieved **R² 0.8632** — **86.3% of grade variance explained** — and **RMSE 1.1547 on the 0–20 grade scale (≈5.8% error)**, documenting experiments and feature importance for reproducibility.

AI Appliance Recommender Chatbot | Python, NLP, LLaMA 3.1, Groq API, Streamlit May' 26

- Built an end-to-end **AI pipeline** converting natural-language queries into **4 structured preferences** — budget, family size, machine type and load type — to drive downstream decision-making.
- Parsed input using **regex and keyword matching** to absorb phrasing variants, then ranked products through a **4-stage pipeline** weighting rating **60%** and capacity **40%** alongside affordability.
- Integrated **LLaMA 3.1 via Groq API** for generated explanations with **Ollama** as a local fallback, and evaluated the system across budget, load-type and invalid-input scenarios.

Efficient Page Replacement Algorithm Simulator | C++, STL, Operating Systems Dec' 26

- Engineered a **C++ simulator** comparing **FIFO, LRU and Optimal** page replacement across configurable reference strings and frame counts.
- Tracked frame state using **vectors, queues, unordered sets and maps**, with the codebase split into modular input, execution, metrics and comparison components.
- Benchmarked side-by-side **page-fault counts and hit ratios** with frame-by-frame indicators, through a console interface for running one or all 3 strategies.

TRAINING

Lovely Professional University — Summer Training Supply Chain Demand Forecasting & Anomaly Detection | Python, Prophet, Scikit-learn, SQLite, Streamlit Jun' 26 – Jul' 26

- Built an end-to-end forecasting pipeline over **421,570 weekly sales records** spanning **45 stores and 99 departments**, cleaned and loaded into a **SQLite** database with a reusable **SQL query layer**.
- Trained **Facebook Prophet** on a **12-week horizon** with yearly seasonality and detected irregular demand via **Isolation Forest** tuned to a **5% expected anomaly rate**, flagging **8 of 143 weeks (5.6%)**.
- Deployed a **4-tab Streamlit dashboard** with live forecasts, confidence intervals, adjustable anomaly sensitivity and SQL-driven analytics; benchmarked at **MAE \$7,057** and **RMSE \$14,540**.

CERTIFICATES

Campus to Corporate: Data & AI Engineer Program | *Lovely Professional University* Jul' 26
Programming Using C++ | *Infosys Springboard* Aug' 25
Computer Programming | *iamneo* May' 25
Essentials of Cyber Security | *Tech Veda* Mar' 25

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology Aug' 24 – Present
Computer Science and Engineering; CGPA: 8.32
Narayana E-Techno School Chennai, Tamil Nadu
Intermediate May' 22 – Apr' 24
Percentage: 74.8%
Narayana Olympiad School Bangalore, Karnataka
Matriculation Apr' 20 – Apr' 22
Percentage: 90%